Week 09: Midterm preparation

Learning outcomes: CLO1, CLO2, CLO3

Lesson

This week is the midterm, which covers Weeks 1–8. Your instructor will give you the time, the rules and the allowed materials. The tasks below are practice only; they are not the exam.

Try each task on your own first, then check your notes. For every query, be ready to explain what one row means, which keys you joined on, how you filtered and how you sorted. For design questions, write the dependencies and explain which rules the schema enforces.

Worked example

Run `example.sql` (see Run a SQL file (`setup/README.md`)) and read the output. Then open `work.sql`.

Tables for this week

Read from the `cambodia` database and make your changes in `student_work`. PK means primary key and FK means foreign key. Every column is `NOT NULL`. The sample rows are the first two rows of each table, sorted by code.

`cambodia.province`

Column	Type	Key	Meaning
`province_pcode`	`CHAR(4)`	PK	Province code
`province_name`	`VARCHAR(100)`		Province name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place

Sample rows:

province_pcode	province_name	area_km2
KH01	Banteay Meanchey	6145.9
KH02	Battambang	11869.0

`cambodia.district`

Column	Type	Key	Meaning
`district_pcode`	`CHAR(6)`	PK	District code
`district_name`	`VARCHAR(100)`		District name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`province_pcode`	`CHAR(4)`	FK	Province code

Sample rows:

district_pcode	district_name	area_km2	province_pcode
KH0102	Mongkol Borei	807.2	KH01
KH0103	Phnum Srok	761.7	KH01

`cambodia.commune`

Column	Type	Key	Meaning
`commune_pcode`	`CHAR(8)`	PK	Commune code
`commune_name`	`VARCHAR(100)`		Commune name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`district_pcode`	`CHAR(6)`	FK	District code

Sample rows:

commune_pcode	commune_name	area_km2	district_pcode
KH010201	Banteay Neang	52.0	KH0102
KH010202	Bat Trang	40.3	KH0102

Join keys: `district.province_pcode = province.province_pcode`; `commune.district_pcode = district.district_pcode`. Join on the codes, not the names.

Metadata used by `example.sql`: `information_schema.COLUMNS` lists every column of every table. The example keeps only `TABLE_SCHEMA = 'cambodia'` and shows `TABLE_NAME`, `COLUMN_NAME`, `COLUMN_TYPE`, `IS_NULLABLE` and `COLUMN_KEY`.

Run `DESCRIBE cambodia.province;` to see a table definition in MySQL. The data guide (`data/README.md`) explains where the data comes from.

Lab tasks

1. Without looking at earlier answers, list province names and areas, sorted by area and then by code.
2. Count the districts in each province, keeping provinces that have no districts.
3. Find the districts larger than the average district in their own province.
4. Draw the ERD of the three tables and explain the primary and foreign keys.
5. Explain how to normalize the Week 8 report. Then delete one row from your own copy of a table inside a transaction and roll it back.

Homework

1. After you get feedback, write your mistakes and the corrected queries in `reflection.md`.
2. Make a one-page study sheet based on the exam rules your instructor gives you.

Capstone sprint

No new sprint this week. Keep your Checkpoint 2 version committed and list your next project tasks in priority order.

Submit

Write your answers in `work.sql` and label each one, for example `-- Lab 3`. Answer the questions in `reflection.md` and save your query output in `results/`. Then commit and push to your repository. See SUBMISSIONS.md for details.